Initial system

$$\begin{cases} \dot{x}_1 = (t+1)\cos(x_3)u_1 \\ \dot{x}_2 = (t+1)\sin(x_3)u_1 \quad x^0 = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix} \\ \dot{x}_3 = u_2 \end{cases}$$

Series

$$x(\theta) = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \eta_0 + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \eta_1 + \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \eta_{00} + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \eta_{01}$$

Basis Lie elements and their coefficients

$$\begin{aligned} \mathcal{F}_1 : g_1 &= \eta_0 & c(g_1) &= \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \\ g_2 &= \eta_1 & c(g_2) &= \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \\ \mathcal{F}_2 : g_3 &= [\eta_0, \eta_1] & c(g_3) &= \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} \end{aligned}$$

Lie elements for principal part

$$\begin{aligned} \mathcal{F}_1 : \ell_1 &= \eta_0 \\ \ell_2 &= \eta_1 \\ \mathcal{F}_2 : \ell_3 &= [\eta_0, \eta_1] \end{aligned}$$

The generating set of the Left ideal

Principal part

$$\begin{aligned} \mathcal{F}_1 : \tilde{\ell}_1 &= \eta_0 \\ \tilde{\ell}_2 &= \eta_1 \\ \mathcal{F}_2 : \tilde{\ell}_3 &= [\eta_0, \eta_1] \end{aligned}$$

Approximating system

$$\begin{cases} \dot{x}_1 = u_1 \\ \dot{x}_2 = u_2 \\ \dot{x}_3 = -x_1 u_2 + x_2 u_1 \end{cases}$$